🍔 QuickBite Crisis Impact Analytics & ML Platform

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📌 Project Overview

The **QuickBite Crisis Impact Analytics & ML Platform** is an end-to-end **Data Engineering + Analytics + Machine Learning** project built entirely on **Databricks**. The goal of this project is to analyze the impact of a crisis period on a food delivery business and provide **data-driven insights** and **predictive intelligence** to support business recovery.

This project was developed as a **self-directed capstone** and demonstrates production-grade practices across data ingestion, transformation, optimization, analytics, ML experimentation, and orchestration.

🎯 Business Problem Statement

During crisis periods (e.g., economic downturns, pandemics), food delivery platforms experience:

- Decline in order volumes
- Increase in cancellations
- SLA breaches due to delivery delays
- Drop in customer ratings
- Customer churn

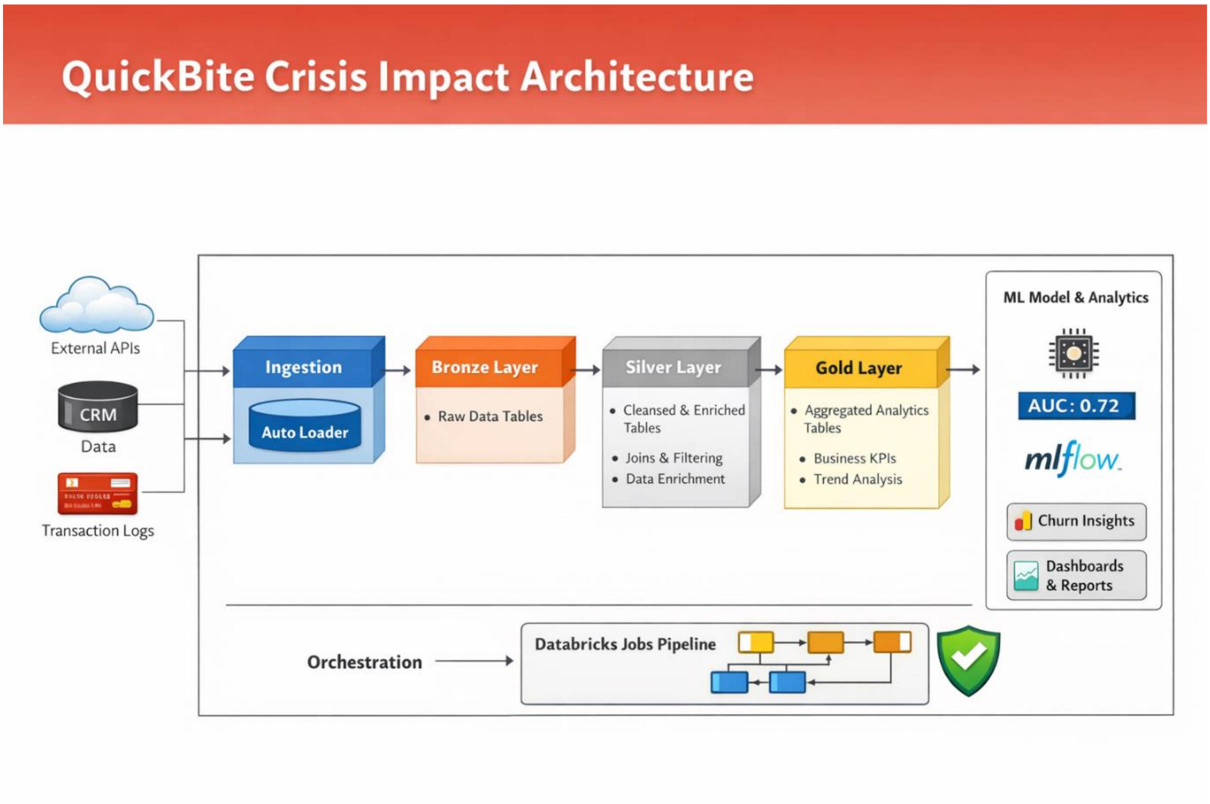
Objective:

- Quantify the business impact of the crisis
- Identify cities and restaurants most affected
- Understand customer behavior changes
- Predict customer churn using ML
- Enable data-backed recovery strategies

🏗️ Architecture Overview

Platform: Databricks (Serverless Compute)

Architecture Pattern: Medallion Architecture



□ Data Architecture (Medallion)

□ Bronze Layer

Purpose: Raw data ingestion without transformation

- Source: Public/Synthetic dataset
- Stored as Delta tables
- Minimal schema enforcement

Tables:

- fact_orders
- fact_ratings
- fact_orderitems
- fact_deliveryperformance
- dim_customers
- dim_deliverypartner

- dim_menuitem
- dim_restaurants

⚪ Silver Layer

Purpose: Data cleaning, enrichment, and business logic

Key transformations:

- Timestamp standardization (to_timestamp)
- City & restaurant enrichment
- Cancellation flags
- Delivery delay calculations
- SLA breach indicators
- Crisis phase classification (Pre-Crisis / Crisis / Recovery)
- Sentiment scoring from customer reviews

Tables:

- orders_enriched
- ratings_enriched
- customer_behavior

Data Quality Checks:

- Null handling
- Schema validation
- Duplicate removal

▯ Gold Layer

Purpose: Analytics-ready business aggregates

Key Gold tables:

- monthly_orders_trend
- city_order_decline
- restaurant_order_decline
- cancellation_trends
- delivery_sla_summary
- monthly_ratings_trend

- revenue_impact
- loyalty_churn
- high_value_customer_decline

Performance Optimizations:

- Delta Lake storage
- OPTIMIZE
- Z-ORDER on frequently filtered columns

📊 Analytics & Insights

The Gold layer enables rich analytics such as:

- Order & revenue trends across crisis phases
- City-wise and restaurant-wise impact analysis
- Cancellation rate comparison
- Delivery SLA degradation
- Rating & sentiment shifts
- Customer loyalty and churn patterns

These tables are directly consumed by **Databricks SQL Dashboards**.

▯ Machine Learning Component

🔍 ML Use Case: Customer Churn Prediction

Problem Framing: Predict whether a customer will churn during a crisis period based on pre-crisis behavior.

Features Used:

- Pre-crisis order count
- Average order value
- Average delivery delay
- Average rating
- Average sentiment score

Model:

- Logistic Regression (chosen for explainability)

Evaluation Metric:

- AUC (Area Under ROC Curve)

Result:

- Achieved AUC ≈ **0.72**

MLflow:

- Experiments tracked with parameters & metrics
- Runs organized by project and use case

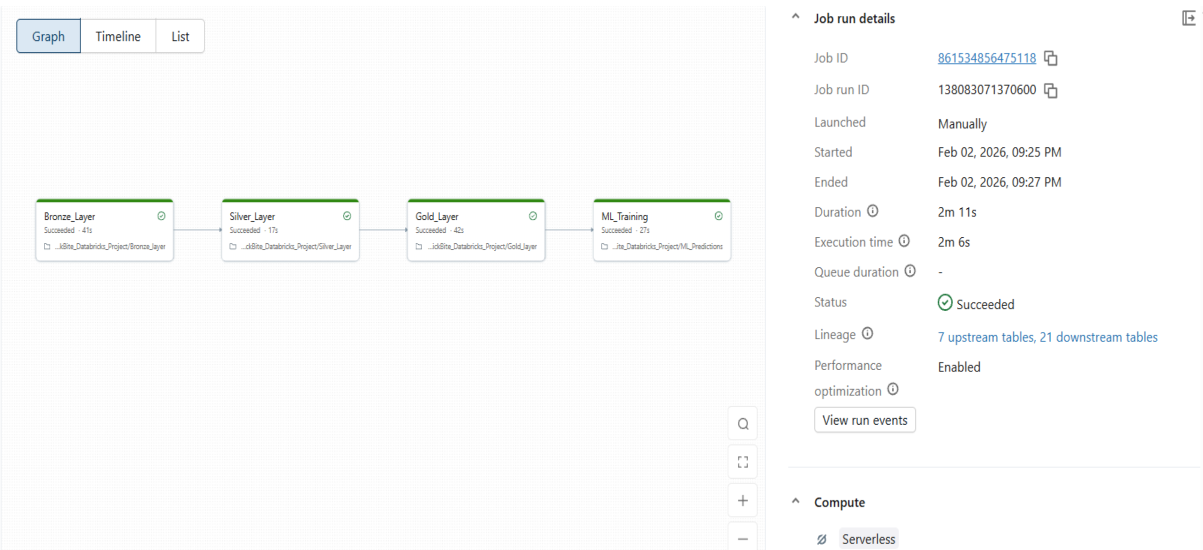
Note: Due to Databricks Serverless + Unity Catalog constraints, model artifact logging was skipped. Metrics and experiment lineage were fully tracked using MLflow.

🔄 Orchestration (Databricks Jobs)

The entire pipeline is automated using **Databricks Workflows (Jobs)**.

Job Flow:

1. Bronze Ingestion
2. Silver Transformations
3. Gold Aggregations
4. Delta Optimization (OPTIMIZE, Z-ORDER)
5. ML Training & Scoring



Key Features:

- Task dependencies
- Retry policies
- Serverless compute

- Repeatable & production-ready execution

🔐 Governance & Best Practices

- Unity Catalog for data organization
- Schema separation (Bronze / Silver / Gold)
- Read/write access separation
- Delta Lake ACID guarantees
- Scalable, cloud-native design

🗂 Project Structure

QuickBite/

├── 01_Bronze_Ingestion

├── 02_Silver_Transformations

├── 03_Gold_Aggregations

├── 04_Optimize_Tables

├── 05_ML_Modeling

├── dashboards/

📌 Key Business Outcomes

- Identified top cities and restaurants impacted during crisis
- Quantified revenue and order volume decline
- Measured delivery SLA degradation
- Detected loyal customer churn patterns
- Built predictive model to target high-risk customers

🛠 Technologies Used

- Databricks (Serverless)
- Apache Spark (PySpark & SQL)
- Delta Lake
- MLflow

- Unity Catalog
- Databricks SQL Dashboards

📎 Submission & Credits

Platform: Databricks

Community Tags:

- Databricks
- Codebasics
- Indian Data Club

🏁 Conclusion

This project demonstrates how Databricks can be used to build a **scalable, governed, and intelligent analytics platform** that combines data engineering, analytics, and machine learning to solve real-world business problems.

☆ *If you found this project insightful, feel free to connect and discuss data engineering & analytics!*